PRODUCT USER MANUAL

For Global Level 3S ODYSSEA Sea Surface Temperature multi-year Product

SST_GLO_PHY_L3S_MY_010_39
Issue: 1.0

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Approval Date : November 2025

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05/13	All	Initial version	E. Autret	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual describing the Copernicus Marine ODYSSEA GLO reprocessed level 3S, which was processed at Ifremer (FR). The manual covers: how it was created, its content, and which data services are available to access it.
The Copernicus Marine ODYSSEA reprocessed L3S product provides a global multi-sensor composite of Sea Surface Temperature (SST) from multiple infra-red instruments onboard polar satellites. The product provides an estimate of the foundation SST which is the SST free of diurnal variability, created by the Ifremer ODYSSEA system using re-processed ESA SST CCI, C3S [RD.1]. The product is available from 1st January 1982 to 1 month before real time on a regular grid spanning from 180° W to 180° E and 80° S to 80° N at 0.1° resolution.
The product is published on the Copernicus Marine dissemination server after automatic and human quality controls. Files downloaded are in NetCDF format and follow CF-1.7 and GHRSST format conventions.
The analysis system is described in the Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SST_GLO_PHY_L3S_MY_010_039
Geographical coverage	Global : 180°W-180°E ; 80°S – 80°N
Variables	• Adjusted_sea_surface_temperature
Product Type	Reprocessing
Available time series	01/01/1982 – 1 month before real time
Temporal resolution	Daily
Update Frequency	1 month
Target delivery time	Monthly, the 25th, at 12 pm (noon) UTC
Horizontal resolution	1/10 degree
Number of vertical levels	Surface only
Format	NetCDF4 (CF1.7, GDS 2.1 conventions)

b) Production System Description

The Copernicus Marine ODYSSEA reprocessed L3S product provides a regional multi-sensor composite of adjusted Sea Surface Temperature from multiple infra-red instruments onboard polar satellites. The satellite derived SST data which are used to reconstruct the daily SST fields, are extracted from the global ESA Climate Change Initiative (CCI) Copernicus Climate Change Service (C3S) dataset. (Embury et al., 2024 [R.1]). This archive includes observations from satellite thermal infra-red measurements from Advanced Very High Resolution Radiometers (AVHRRs), Along-Track Scanning Radiometers (ATSRs), the Sea and Land Surface Temperature Radiometer (SLSTR) on-board SENTINEL-3 and from microwave observations as described in [R.1]. In the L3S product, the absence of valid SST values is mainly due to the presence of clouds. Only observations with quality level values greater than 4 (out of 5, corresponding to acceptable and best quality measurements in GHRSST format convention) have been included in the analysis. In order to avoid the use of data affected by diurnal warming, only night-time observations have been included.

The L3S product is derived from a collection of sea surface temperature (SST) data acquired by various instruments, resulting in multiple SST observations for each pixel on a given day. Due to differences in the spectral channels used and variations in instrument sensitivity, SST measurements from different sensors may exhibit slight discrepancies. These differences typically occur at a spatial scale of approximately 100 km. Directly aggregating data from these diverse sources without accounting for such biases could introduce artifacts, including discontinuities and noise, into the L3S SST products.

To ensure the consistency of the daily L3S dataset, intercalibration of the various data sources is required. This process involves estimating a daily correction to be applied to SST values from each individual satellite source. First, a daily reference SST field is constructed by computing the median of all available observations. Then, for each satellite data source, a large-scale bias field is derived by calculating the difference between the observations and the daily reference, with the anomaly smoothed using a 70 km Gaussian filter. The estimated correction is subsequently subtracted from the SST values of each satellite source, producing the L3S *adjusted_sea_surface_temperature* variable.

The daily individual sensor composite files are then merged into a combined multi-sensor composite file (L3S) at the final product resolution. The merging process takes into account a hierarchy of sensors (from best to worst) (again, estimated from comparisons of each source with in situ measurements). Any given cell of the final grid will contain a value from one source only, which is means the best is selected. The used source is reported in the *sources_of_sst* variable of the created L3S.

The ODYSSEA processing chain was developed and is maintained by the Laboratory for Ocean Physics and Satellite remote sensing (LOPS) at the Institut Français de Recherche pour l'Exploitation de la Mer (Ifremer, http://www.ifremer.fr/). It is run operationally for CMEMS SST TAC production at Ifremer marine data operational service (*SISMER* : Systèmes d'informations scientifiques pour la Mer).

GLO (180°W-180°E; 80°S – 62°N): 1/10°, regular grid, 1800 x 3600 x 1
with 1time step per file
Adjusted SST provided over the Global ocean

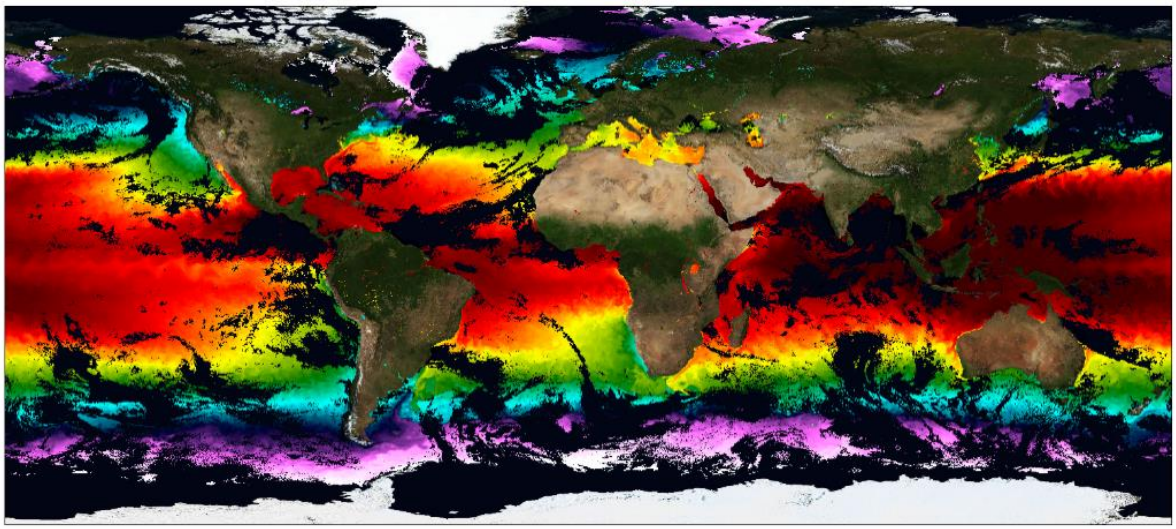

Figure 1: GLO area illustrated by a global daily L3S

Vertical grid	Surface only (20 cm)
System Name	ODYSSEA-REP version 3.0
Horizontal Resolution	1/10°
Input observations	SST satellite data: A collection of L3U and L3C data from various sources (ESA SST CCI, C3S and EUMETSAT; see quality information document for full details). All data are provided with Sensor Specific Error Statistics (SSES_bias and SSES_std variables) that give an estimate of the systematic and random errors at pixel level, and quality level flags. A bias correction is performed using ESA SST CCI / C3S data [RD.1] as reference.

c) Dataset details

Product Lines	SST_GLO_PHY_L3S_MY_010_039	
Geographical coverage	Global	
cmems_obs-sst_glo_phy_my_l3s_P1D-m		
Product Type	Multi-year – L3S	
Update frequency	Montly	
Available time series	-1M - up to real-time01/01/1982 to -1 month before real time	
Temporal resolution	Daily mean	

Dataset : cmems-obs-sst-glo-phy-my-l3s_P1D-m contains daily satellite observations of foundation sea surface temperature.
adjusted_sea_surface_temperature [K] sea surface temperature corrected with sensor bias error, skin to subskin adjustment and cross-sensor intercalibration, averaged to grid resolution. **The recommended SST value to use**. sea_surface_foundation_temperature
sea_surface_temperature [K] Uncorrected raw sea surface temperature as found in the input source SST products merged here, averaged to grid resolution. sea_surface_foundation_temperature
sst_dtime [second] time difference of the observation value from the file reference time
quality_level quality level of the SST value (copied from the source input). These are the overall quality indicators and are used for all GHRSST SSTs
sses_bias [K] bias estimate of the uncorrected sea surface temperature value, as provided in the source input.
sses_standard_deviation [K] error of the uncorrected sea surface temperature value, as provided in the source input.
sources_of_sst product origin of the SST value
bias_to_reference_sst [K] bias error estimated to correct the original SST value.

d) Additional information on some parameters

N/A

e) Processing information

Product disseminating schedule :

Type of product	Observation (base time)	Product temporal coverage (hours)	Product disseminating scheduled time
Reprocessing	12 UTC	-12 to +12	Monthly

f) Catalogue updating

The catalogue data cover 1982 to present minus one month. It is updated monthly.

g) Time averaging

Dataset frequency	Time averaging
daily	daily means over a day (centered at noon).

h) Product use and impact

This product is meant for a large range of applications requiring foundation (not affected by diurnal cycle) high resolution sea surface temperature at global scale. The L3S provides the satellite observations, contrary to Level 4 datasets whose gap free sea surface temperature comes at the cost of optimal interpolation which tends to smooth the observations and is prone to errors in the estimation of the sea surface temperature over areas with persistent cloud coverage. As a consequence, the L3S provides higher and sharper resolution of ocean features. Users that want to rely on true observations rather than interpolated observations should use the L3S.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Example :

20230101000.nc000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-ATL_MY_005_adjusted-v03.0-f01.0.nc

4.1 FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [MB]
cmems_obs-sst_glo_phy_my_l3s_P1D-m	<date>000000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-GLO_MY_010_adjusted-v03.0-fv01.0.nc	4

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-026-038-039.pdf

[R.1] Embury, O., Merchant, C. J., Good, S. A., Rayner, N. A., Høyer, J. L., Atkinson, C., ... & Donlon, C. (2024). Satellite-based time-series of sea-surface temperature since 1980 for climate applications. *Scientific Data*, *11*(1), 326.

6) ANNEX

An example header of a high resolution L3S netCDF file (generated using ncdump) is given below

```
Netcdf \20230101000000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-GLO_MY_010_adjusted-v03.0-f01.0 {
dimensions:
        time = 1 ;
        lat = 1800 ;
        lon = 3600 ;
variables:
        double time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
        double lat(lat) ;
                lat:long_name = "latitude" ;
                lat:standard_name = "latitude" ;
                lat:axis = "Y" ;
                lat:units = "degrees_north" ;
        double lon(lon) ;
                lon:long_name = "longitude" ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
        int sst_dtime(time, lat, lon) ;
                sst_dtime:_FillValue = -2147483648 ;
                sst_dtime:long_name = "time difference from reference time" ;
                sst_dtime:units = "seconds" ;
                sst_dtime:add_offset = 0. ;
                sst_dtime:scale_factor = 1. ;
                sst_dtime:valid_min = -2147483647 ;
                sst_dtime:valid_max = 2147483647 ;
        short sea_surface_temperature(time, lat, lon) ;
                sea_surface_temperature:_FillValue = -32768s ;
                sea_surface_temperature:long_name = "sea surface temperature" ;
```

```
                sea_surface_temperature:standard_name =
"sea_surface_foundation_temperature" ;
                sea_surface_temperature:type = "foundation" ;
                sea_surface_temperature:units = "kelvin" ;
                sea_surface_temperature:add_offset = 273.15 ;
                sea_surface_temperature:scale_factor = 0.01 ;
                sea_surface_temperature:valid_min = -300s ;
                sea_surface_temperature:valid_max = 4500s ;
                sea_surface_temperature:authority = "CF-1.7" ;
        short adjusted_sea_surface_temperature(time, lat, lon) ;
                adjusted_sea_surface_temperature:_FillValue = -32768s ;
                adjusted_sea_surface_temperature:long_name = "sea surface temperature" ;
                adjusted_sea_surface_temperature:standard_name =
"sea_surface_temperature" ;
                adjusted_sea_surface_temperature:type = "foundation" ;
                adjusted_sea_surface_temperature:units = "kelvin" ;
                adjusted_sea_surface_temperature:add_offset = 273.15 ;
                adjusted_sea_surface_temperature:scale_factor = 0.01 ;
                adjusted_sea_surface_temperature:valid_min = -300s ;
                adjusted_sea_surface_temperature:valid_max = 4500s ;
                adjusted_sea_surface_temperature:authority = "CF-1.7" ;
        byte sses_bias(time, lat, lon) ;
                sses_bias:_FillValue = -128b ;
                sses_bias:long_name = "SSES bias estimate" ;
                sses_bias:units = "kelvin" ;
                sses_bias:add_offset = 0. ;
                sses_bias:scale_factor = 0.01 ;
                sses_bias:valid_min = -127b ;
                sses_bias:valid_max = 127b ;
                sses_bias:comment = "Populated with zeroes as input CCI and C3S data are
already adjusted and unbiased" ;
        short sses_standard_deviation(time, lat, lon) ;
                sses_standard_deviation:_FillValue = -32768s ;
                sses_standard_deviation:long_name = "Total uncertainty in each
sea_surface_temperature data point" ;
                sses_standard_deviation:units = "kelvin" ;
                sses_standard_deviation:add_offset = 0. ;
                sses_standard_deviation:scale_factor = 0.001 ;
                sses_standard_deviation:valid_min = 0s ;
                sses_standard_deviation:valid_max = 5000s ;
                sses_standard_deviation:comment = "Total uncertainty in each
sea_surface_temperature data point from CCI and C3S input data" ;
        byte quality_level(time, lat, lon) ;
                quality_level:_FillValue = -128b ;
                quality_level:long_name = "quality level value" ;
                quality_level:valid_min = 0b ;
                quality_level:valid_max = 5b ;
                quality_level:flag_meanings = "no_data bad_data worst_quality low_quality
acceptable_quality best_quality" ;
                quality_level:flag_values = 0b, 1b, 2b, 3b, 4b, 5b ;
                quality_level:comment = "These are overall quality indicators and are used for
all GHRSST SSTs" ;
        byte sources_of_sst(time, lat, lon) ;
                sources_of_sst:_FillValue = -128b ;
```

```
                sources_of_sst:long_name = "source of sea surface temperature" ;
                sources_of_sst:valid_min = 1b ;
                sources_of_sst:valid_max = 19b ;
                sources_of_sst:flag_meanings = "AVHRR_NOAA_07 AVHRR_NOAA_09
AVHRR_NOAA_11 AVHRR_NOAA_12 AVHRR_NOAA_14 AVHRR_NOAA_15
AVHRR_NOAA_16 AVHRR_NOAA_17 AVHRR_NOAA_18 AVHRR_NOAA_19 ATSR1_ERS1
ATSR2_ERS2 AATSR_ENVISAT AVHRR_METOP_A AVHRR_METOP_B
SLSTR_SENTINEL3A SLSTR_SENTINEL3B AMRSE_AQUA AMSR2_GCOMW1" ;
                sources_of_sst:flag_values = 1b, 2b, 3b, 4b, 5b, 6b, 7b, 8b, 9b, 10b, 11b, 12b,
13b, 14b, 15b, 16b, 17b, 18b, 19b ;
        short bias_to_reference_sst(time, lat, lon) ;
                bias_to_reference_sst:_FillValue = -32768s ;
                bias_to_reference_sst:long_name = "bias to the reference sea surface
temperature used for cross-calibration" ;
                bias_to_reference_sst:units = "kelvin" ;
                bias_to_reference_sst:add_offset = 0. ;
                bias_to_reference_sst:scale_factor = 0.01 ;
                bias_to_reference_sst:valid_min = -32767s ;
                bias_to_reference_sst:valid_max = 32767s ;

// global attributes:
                :Conventions = "CF-1.7,ACDD-1.3,ISO 8601" ;
                :standard_name_vocabulary = "NCDF Climate and Forecast (CF) Metadata
Convention version 1.7" ;
                :naming_authority = "org.ghrsst" ;
                :netcdf_version_id = "4.6.1 of Sep  8 2018 17:21:01" ;
                :title = "Merged collation of sea surface temperature from multiple satellite
sources" ;
                :id = "IFR-L3S-GLO-ODYSSEA" ;
                :cmems_product_id = "SST_GLO_PHY_L3S_MY_010_038" ;
                :summary = "Multi-sensor composite on a global 0.1 degree resolution " ;
                :references = "Product User Manual for Global Level 3 ODYSSEA Sea Surface
Temperature multi-year Product, Nov 2025" ;
                :metadata_link = "" ;
                :processing_level = "L3S" ;
                :keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
                :keywords_vocabulary = "NASA Global Change Master Directory (GCMD)
Science Keywords" ;
                :project = "Copernicus - Marine environment monitoring service (CMEMS)" ;
                :program = "GHRSST, CMEMS" ;
                :institution = "Institut Francais de Recherche pour l Exploitation de la mer /
CERSAT" ;
                :institution_abbreviation = "Ifremer/Cersat" ;
                :licence = "These data are available free of charge under the CMEMS data
policy, refer to http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence/" ;
                :acknowledgment = "Please acknowledge the use of these data with the
following statement: these data were obtained from Copernicus Marine Service and produced
by the Centre de Recherche et d Exploitation Satellitaire (CERSAT), at Ifremer, Plouzane
(France" ;
                :contact = "jean.francois.piolle@ifremer.fr;emmanuelle.autret@ifremer.fr" ;
                :technical_support_contact = "cersat@ifremer.fr" ;
                :scientific_support_contact = "emmanuelle.autret@ifremer.fr;jfpiolle@ifremer.fr"
;
```

:creator_email = "cersat@ifremer.fr" ;
:creator_type = "institution" ;
:creator_institution = "Ifremer / CERSAT" ;
:creator_name = "CERSAT" ;
:creator_url = "http://cersat.ifremer.fr" ;
:publisher_email = "servicedesk.cmems@mercator-ocean.eu" ;
:publisher_institution = "CMEMS" ;
:publisher_name = "CMEMS" ;
:publisher_url = "marine.copernicus.eu" ;
:format_version = "GHRSST GDS v2.1" ;
:gds_version_id = "2.1" ;
:processing_software = "ODYSSEA REP 3.0" ;
:source = "Odyssea-RepL3 Processor" ;
:source_version = "1.0" ;
:geospatial_bounds = "POLYGON ((-21.0 9.0, 13.0 9.0, 13.0 62.0, -21.0 62.0, -21.0 9.0))" ;
:geospatial_bounds_crs = "WGS84" ;
:geospatial_bounds_vertical_crs = "EPSG:5831" ;
:geospatial_lat_min = "-79.50" ;
:geospatial_lat_max = "79.50" ;
:geospatial_lon_min = "-179.50" ;
:geospatial_lon_max = "179.5" ;
:geospatial_lat_units = "degrees_north" ;
:geospatial_lat_resolution = "0.10" ;
:geospatial_lon_units = "degrees_east" ;
:geospatial_lon_resolution = "0.10" ;
:time_coverage_start = "20230101T000000Z" ;
:time_coverage_end = "20230102T000000Z" ;
:time_coverage_resolution = "P1D" ;
:spatial_resolution = "0.1 degree" ;
:temporal_resolution = "daily" ;
:cdm_data_type = "grid" ;
:source_data = "AVHRR_NOAA_07 AVHRR_NOAA_09 AVHRR_NOAA_11 AVHRR_NOAA_12 AVHRR_NOAA_14 AVHRR_NOAA_15 AVHRR_NOAA_16 AVHRR_NOAA_17 AVHRR_NOAA_18 AVHRR_NOAA_19 ATSR1_ERS1 ATSR2_ERS2 AATSR_ENVISAT AVHRR_METOP_A AVHRR_METOP_B SLSTR_SENTINEL3A SLSTR_SENTINEL3B AMSRE_AQUA AMSR2_GCOMW1" ;
:instrument = "AVHRR/2 AVHRR/2 AVHRR/2 AVHRR/2 AVHRR/2 AVHRR/3 AVHRR/3 AVHRR/3 AVHRR/3 AVHRR/3 ATSR ATSR AATSR AVHRR/3 AVHRR/3 SLSTR SLSTR AMSRE AMSR2" ;
:platform_type = "low_earth_orbit_satellite low_earth_orbit_satellite" ;
:platform_vocabulary = "CEOS" ;
:product_version = "1.0" ;
:history = "collated data file originally produced by Ifremer/Cersat with Odyssea Rep 3.0 processor" ;
:uuid = "" ;
:date_created = "20241016T163348Z" ;
:file_quality_level = "" ;

```
                :comment = "WARNING:Some applications are unable to properly handle
signed byte values. If values are encountered > 127, please substract 256 from this reported
value" ;
                :platform = "MetOpB,Sentinel3A,Sentinel3B" ;
}
```